

PHYS 105 – Practical Physics: Mechanics and Thermal Physics

Uncertainty analysis

How good is this number, and how do I show it?

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Reading: OpenStax *University Physics* Vol. 1, §1.6; Taylor, *An Introduction to Error Analysis*, Ch. 1–4 and 8; Hughes and Hase, *Measurements and Their Uncertainties*, Ch. 1–6

By the end of this lecture, you should be able to:

1. Report a result as (value $\pm$ uncertainty) unit, correctly rounded, with its percent uncertainty.
2. Distinguish accuracy, precision, random and systematic uncertainty, and classify a source as Type A or Type B.
3. Calculate the mean, standard deviation and standard error of repeated readings.
4. Propagate uncertainties through a formula and find the dominant source.
5. Fit a straight line, interpret its residuals and χ^2 , and report slope and intercept.
6. Build an uncertainty budget and test agreement with an accepted value.

Three contacts:

1 UNDERSTAND

What is a result, and where does uncertainty come from?

2 QUANTIFY

How do repeated readings and formulas give a number with an uncertainty?

3 APPLY

Graphs, budgets, comparison with theory, and the report.

Practice: concept check $\rightarrow$ lecturer example
 $\rightarrow$ independent $\rightarrow$ application

Not marked. Answer individually, then we review.

1. How many significant figures are in 0.0450, in 4.50×10^3 and in 1020?
2. What percentage is 0.5 of 40?
3. Round 7.846 to two decimal places and to two significant figures.
4. A ruler is marked in millimetres. What is the smallest length you can read directly?
5. The slope of a straight-line graph of y against x has what units?
6. Differentiate $f = x^2$ and $f = \sin x$.

1. 3, 3, 4. Leading zeros are placeholders; the trailing zero in 0.0450 is significant.
2. $0.5/40 = 1.25\%$.
3. 7.85; 7.8.
4. 1 mm, with the last digit estimated by eye.
5. Units of y divided by units of x .
6. $2x$; $\cos x$ (with x in radians).

Contact 1: Understand

*What is a measurement result, and
where does uncertainty come from?*

Two students measure the length of the same bench

Student X reports: “the bench is 1.2 m long.”

Student Y reports: “the bench is 1.213 ± 0.002 m long.”

Predict first

Which report is more useful to a third person who must cut a shelf to fit? What does Y tell you that X does not? Is either report certainly *true*?

- X gives one number and no sense of how far to trust it.
- Y gives a best value and a range. The shelf can be cut to about 1.21 m with a known margin.
- Neither report is certain: every measurement has an uncertainty, and **saying so is part of the result.**

A measurement is not complete without its uncertainty

Rule: the form of a measurement result

A result has three parts: a **best estimate**, an **uncertainty** and a **unit**:

$$x = (x_{\text{best}} \pm \delta x) \text{ unit}$$

δx is the **uncertainty**: the size of the range, around the best estimate, within which the true value is expected to lie with reasonable confidence.

Physical meaning

“Uncertainty” is not “mistake”. A mistake (reading the wrong scale, writing the wrong digit) can be avoided. Uncertainty cannot be avoided; it can only be *estimated* and *reduced*. It is the honest statement of what the measurement can and cannot tell us.

Notation follows OpenStax §1.6 ($A \pm \delta A$). δx is one *standard uncertainty* (later slides).

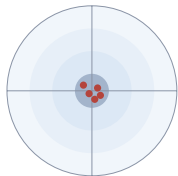
Vocabulary: four words that are often confused

Term	Meaning	Needs a reference value?
Accuracy	How close a measurement is to the accepted reference value.	Yes
Precision	How closely repeated measurements, under the same conditions, agree with each other.	No
Uncertainty δx	A quantitative measure of the spread or doubt in the result. Related to precision.	No
Discrepancy	The difference between the measured value and the accepted value: $x_{\text{best}} - x_{\text{acc}}$. Related to accuracy.	Yes

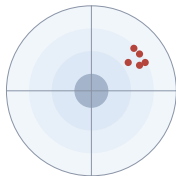
Caution

In everyday speech “error” means a mistake. In practical physics the *difference from the accepted value* is the discrepancy, and the *doubt in the result* is the uncertainty. Neither is a mistake.

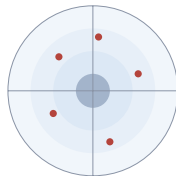
Sketch first: accuracy and precision



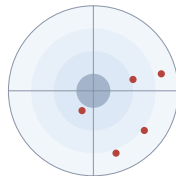
(a) **precise, accurate**
small spread, centred



(b) **precise, not accurate**
small spread, offset



(c) **accurate, not precise**
large spread, centred



(d) **neither**
large spread, offset

Predict first

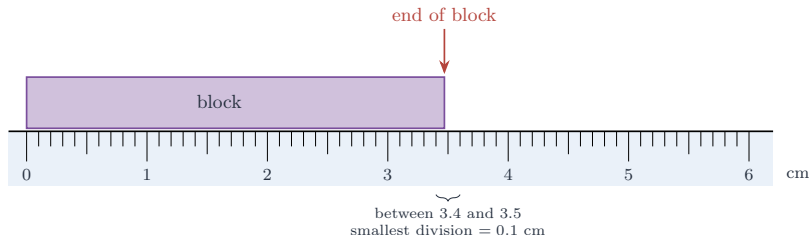
Each dot is one measurement of the same quantity; the centre is the accepted value. Before reading the labels: which panel has small uncertainty and small discrepancy? Which has small uncertainty but large discrepancy? What kind of problem would produce panel (b)?

- Panel (a): the ideal. Both small.
- Panel (b): the scatter is small but everything is shifted. A *consistent* cause, such as a worn ruler end, a clock that runs fast, or a thermometer that reads 1°C high. Repeating more times will **not** remove it.
- Panel (c): scatter large but centred. Averaging helps (Contact 2).
- Panel (d): a scatter problem and a shift.

Why it matters

Precision can be estimated from the data alone. Accuracy cannot: it needs an independent reference value or a different method. So **high precision does not guarantee accuracy**.

Reading an instrument



Which value should be written down?

The end of the block lies between 3.4 and 3.5 cm. We read 3.47 cm by estimating where it falls between two marks. The last digit is an *estimate*. How uncertain is it?

TOOLKIT 1 – Reading an instrument

Resolution is the smallest change the instrument can show. It sets a lower limit on the uncertainty of one reading.

1. **Analogue scale:** read to a fraction of the smallest division. A safe default is a reading limit of **half the smallest division**. State a smaller value only if you can justify it.
2. **Digital display:** the limit is at least **one unit in the last digit**, unless the manufacturer states otherwise.
3. **Write** the unit and every digit the instrument gives, including a final zero.
4. **Zero:** check the reading with nothing being measured (jaws closed, balance empty).

WATCH OUT: the reading limit is only one source. It is not the whole uncertainty of the measurement.

Concept check: how uncertain is each reading?

Assign a reading limit and write each result in full

1. Metre rule, smallest division 1 mm, block end at 3.47 cm.
2. Digital balance displays 152.4 g.
3. Stopwatch displays 15.82 s (resolution 0.01 s).
4. Analogue thermometer, 1 °C divisions, mercury level halfway between 23 and 24 °C.

Predict first

Before looking at the next slide: in which of the four is the instrument's resolution *not* the main concern in practice?

1. Half of 1 mm is 0.5 mm = 0.05 cm: (3.47 ± 0.05) cm. The second decimal is already uncertain, so do not quote 3.470.
2. Last digit is 0.1 g: (152.4 ± 0.1) g.
3. Last digit is 0.01 s: (15.82 ± 0.01) s from the display. But a hand-started stopwatch is limited by *reaction time*, typically 0.1 to 0.2 s, far larger than the display. This is the answer to the prediction.
4. Half of 1 °C is 0.5 °C: (23.5 ± 0.5) °C.

Lesson

The resolution is a *minimum* uncertainty. The real uncertainty is set by the largest source present.

Defining the vocabulary of vernier calipers and the micrometer

Quantity	What it is	What it measures	Unit	Representation
Least count (LC)	the smallest change the instrument can show	the resolution	mm	vernier step; one thimble division
Vernier scale	a sliding scale of N divisions spanning $N - 1$ main divisions	fractions of a main division	mm	the line that coincides
Pitch	the distance the screw advances in one turn	the coarse step of the micrometer	mm	0.5 mm per turn
Zero error x_0	the reading with the jaws closed	a constant offset	mm	signed number
Parallax	the apparent shift of a mark seen from the side	an observer error	—	eye not square to the scale

Interpret: the least count answers *what is the smallest step I can read?* It is a resolution, not yet the uncertainty.

Concept check: calipers and micrometers before the formulas

Decide first

1. Is the least count the same as the uncertainty of a reading?
2. Can you read a micrometer to 0.001 mm?
3. Closed jaws read +0.03 mm. Does the zero error make a measured 5.47 mm larger or smaller?
4. Does viewing a scale from the side matter for a digital caliper?

Concept check: calipers and micrometers before the formulas

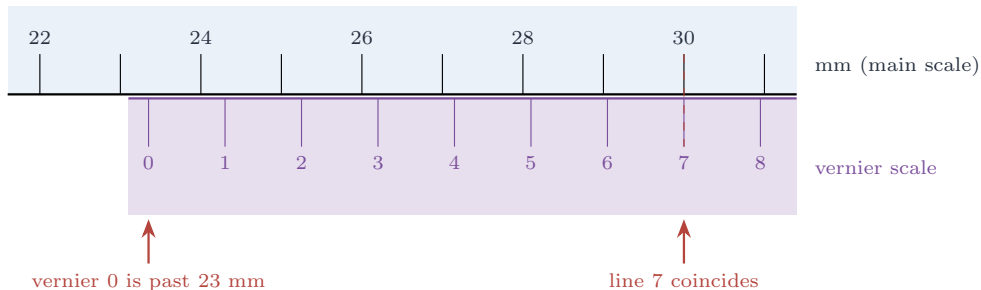
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2. Can you read a micrometer to 0.001 mm?
3. Closed jaws read +0.03 mm. Does the zero error make a measured 5.47 mm larger or smaller?
4. Does viewing a scale from the side matter for a digital caliper?

Answers

1. **No.** It is the resolution; the uncertainty also includes judgement, zero error and technique.
2. **No.** One thimble division is 0.01 mm; you may estimate between lines at most to half of it.
3. **Smaller:** the corrected value is $5.47 - 0.03 = 5.44$ mm.
4. **No** for the reading, but a tilted or loosely clamped jaw still matters.

The vernier caliper: how to read it



Reading steps

(1) Main scale: the last mark *before* the vernier zero, here 23 mm. (2) Vernier: the line that coincides exactly with a main-scale mark, here line 7. (3) Add: $23 + 7 \times \text{LC}$.

EQUATION – vernier caliper

$$\text{LC} = \frac{\text{one main division}}{N}, \quad x = x_{\text{main}} + n \text{LC}$$

Meaning. each vernier division is shorter than a main division by one least count.

Variables. N vernier divisions; n the coincident line.

Units. mm.

Interpretation. $N = 20$ gives 0.05 mm; $N = 10$ gives 0.1 mm; $N = 50$ gives 0.02 mm.

Example. $23 + 7 \times 0.05 = 23.35$ mm.

Predict first

A vernier has $N = 10$ divisions on 9 mm. Predict its least count before using the equation.

Worked example: a vernier reading with zero error

IDENTIFY. Calipers with $N = 20$ divisions on 19 mm. On a rod: main scale 23 mm, line 7 coincides. With the jaws closed the zero reads +0.05 mm (line 1). Required: the corrected diameter and its uncertainty.

SET UP. $LC = 1/20 = 0.05$ mm. A coincidence is judged to one line, so the reading limit is $a = LC$ and $u = a/\sqrt{3}$ for each reading.

Prediction: the corrected value is a little smaller than the raw reading.

EXECUTE.

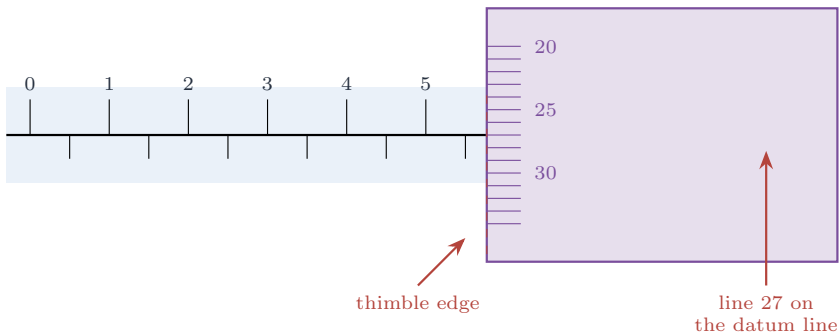
Raw: $23 + 7(0.05) = 23.35$ mm.

Corrected: $23.35 - 0.05 = 23.30$ mm.

$$u = \sqrt{2} \frac{0.05}{\sqrt{3}} = 0.041 \text{ mm.}$$

EVALUATE. $d = (23.30 \pm 0.04)$ mm (0.17%). Smaller than the raw value, as predicted. The zero error contributes as much uncertainty as the reading itself, so it must not be skipped.

The micrometer screw gauge: how to read it



Reading steps

(1) Sleeve: whole millimetres, here 5. (2) The half-millimetre mark below the datum line is visible: add 0.5. (3) Thimble line on the datum line: 27, so add $27 \times \text{LC}$.

EQUATION – micrometer

$$LC = \frac{\text{pitch}}{N}, \quad x = x_{\text{sleeve}} + n LC$$

Meaning. one turn advances the screw by the pitch; the thimble splits it into N steps.

Variables. pitch 0.5 mm; $N = 50$; n the thimble line.

Units. mm.

Interpretation. $LC = 0.01$ mm; the half-mm mark is easily forgotten.

Example. $5.5 + 27 \times 0.01 = 5.77$ mm.

Predict first

If the thimble had 100 divisions for the same pitch, predict the least count.

Worked example: a micrometer reading with a negative zero error

IDENTIFY. A micrometer (pitch 0.5 mm, 50 divisions) reads 5.77 mm on a wire. With the jaws closed the thimble stops at line 47 below the datum line, that is -3 divisions. Required: the corrected diameter and its uncertainty.

SET UP. $x_0 = -0.03$ mm. A thimble line can be judged to about half a division, so $a = 0.005$ mm and $u = a/\sqrt{3}$ for each reading. *Prediction:* subtracting a *negative* zero error makes the wire *larger*.

EXECUTE.

$$x = 5.77 - (-0.03) = 5.80 \text{ mm.}$$

$$u = \sqrt{2} \frac{0.005}{\sqrt{3}} = 0.0041 \text{ mm.}$$

EVALUATE. $d = (5.800 \pm 0.004) \text{ mm.}$

Larger than the raw reading, as predicted. The uncertainty is only 0.07%, so clamping force and a dirty anvil (next slides) may be larger than 0.004 mm: check them.

EQUATION – zero correction

$$x = x_{\text{obs}} - x_0$$

Meaning. remove the constant offset of the instrument.

Variables. x_{obs} the reading; x_0 the closed-jaw reading, with sign.

Units. mm.

Interpretation. positive x_0 : readings too large; negative: too small.

Rule: which reading limit a to use

Full least count: the reading is by coincidence (vernier) or by a digital last digit.

Half least count: a fine scale line can be judged between marks (micrometer thimble, analogue dial).

Then $u = a/\sqrt{3}$. Use the manufacturer's accuracy if it is larger.

Instrument	Least count	Reading limit a	$u = a/\sqrt{3}$
Vernier, 10 divisions	0.1 mm	0.1 mm	0.058 mm
Vernier, 20 divisions	0.05 mm	0.05 mm	0.029 mm
Vernier, 50 divisions	0.02 mm	0.02 mm	0.012 mm
Micrometer	0.01 mm	0.005 mm (half)	0.0029 mm

Other instrument-specific errors

Source	Kind	Effect	Remedy
Parallax (eye not square to the scale)	systematic, per observer	a coincidence is misjudged by a line	look straight on; read twice
Too much clamping force	random or systematic	a soft object is squeezed; micrometer reads small	use the ratchet; same force every time
Backlash in the screw	systematic	different readings turning in and out	always close in the same direction
Dirt or burrs on jaws and anvil	systematic	shifts the zero	clean; re-check zero before and after
Jaws not parallel to the surface	random	overestimates the width	seat the object squarely; repeat at several places
Worn or bent jaws, drifting zero	systematic	zero error changes	recheck x_0 during the session
Warm hands, thermal expansion	systematic	small length change	hold by the insulated frame

Repeat at several points: the scatter then includes the non-uniformity of the object.

IDENTIFY. The length of the pendulum is $L = \ell + d/2$: the string from the support to the top of the bob, $\ell = 0.7880$ m, plus the bob radius. The bob diameter by vernier calipers (LC 0.05 mm, zero error 0) is $d = 24.05$ mm. How much does the caliper add to δL ?

SET UP. $u_d = 0.05/\sqrt{3} = 0.029$ mm; L is a sum: absolute uncertainties. *Prediction:* negligible beside the 2 mm of the lecture.

EXECUTE. $d/2 = 12.025$ mm, so $L = 788.0 + 12.025 = 800.0$ mm.

$$u_{d/2} = 0.0144 \text{ mm.}$$

$$\delta L = \sqrt{2.0^2 + 0.0144^2} = 2.0 \text{ mm.}$$

EVALUATE. $L = (0.8000 \pm 0.0020)$ m: the caliper is irrelevant. The length budget is dominated by locating the string end and the centre of mass. Measuring the bob with a micrometer would change nothing: *spend effort where the budget says.*

Common mistakes and the correct habit

Mistake

Taking the least count as the uncertainty.

Forgetting the half-millimetre mark on the sleeve.

Using 0.1 mm per line on a 0.05 mm vernier.

Reading the line next to the coincident one.

Adding the zero error instead of subtracting it.

Writing 5.8 mm for 5.800 mm.

Clamping hard or skipping the ratchet.

Instead

Use the reading limit a (full or half LC) and $u = a/\sqrt{3}$; add other sources.

Check whether it is visible: it adds 0.5 mm.

Compute LC from the scale first.

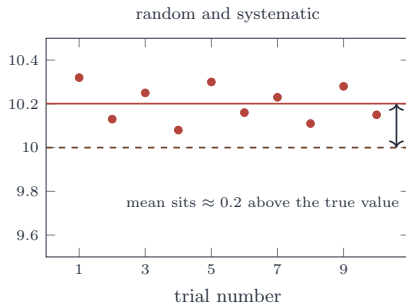
Find the line that is exactly aligned, from the front.

$x = x_{\text{obs}} - x_0$, with the sign of x_0 .

Quote every digit the instrument gives.

Use the ratchet; same force each time.

Sketch first: do repeated readings remove the uncertainty?



Predict first

Ten readings of the same quantity are shown in two situations. If we average each set of ten, where does each average land relative to the true value? If we take 1000 readings instead of ten, what changes in each panel, and what does not?

Rule: two kinds of uncertainty

Random: unpredictable, equally likely to be above or below the true value, changes from reading to reading. *Averaging reduces it.*

Systematic: shifts every reading the same way (same size and direction). *Averaging does not reduce it.* It is found by calibration, by comparison with another method, or by checking the instrument, and is then corrected.

Typical random sources

Reaction time when starting and stopping

Judging a pointer or the end of an object

Electrical noise, air draughts

Small variations in the object itself

Typical systematic sources

Zero error (micrometer closed reads +0.03 mm)

Instrument out of calibration

Parallax from always viewing from one side

A false assumption in the method

Rule: how an uncertainty is evaluated (the GUM classification)

Type A: evaluated from the *statistics of repeated measurements* (the scatter of the readings; Contact 2).

Type B: evaluated by *any other means*: instrument resolution, a manufacturer's specification, a calibration certificate, a published value, or reasoned judgement.

Do not confuse the two classifications

Random against systematic describes the *physical behaviour* of the source. Type A against Type B describes the *method of evaluation*. A random source is usually Type A. A systematic source can only be Type B, because repeating does not reveal it. A resolution limit is Type B but behaves randomly.

Source: JCGM 100:2008 (GUM); NIST TN 1297. First-year laboratories often use only “random” and “systematic”.

Turning a reading limit into a standard uncertainty

Rule: from a limit $\pm a$ to a standard uncertainty

If a quantity is known only to lie within $\pm a$ of the reading, with no value in that range more likely than another (a *rectangular* distribution), the standard uncertainty is

$$u = \frac{a}{\sqrt{3}} \approx 0.58 a.$$

Instrument	Limit a	$u = a/\sqrt{3}$
Metre rule, 1 mm divisions	0.5 mm	0.29 mm
Digital stopwatch, 0.01 s	0.01 s (one unit in the last digit)	0.0058 s
Micrometer, 0.01 mm	0.005 mm	0.0029 mm

Check what your laboratory manual asks

Some laboratories quote the limit a itself as the reading uncertainty. That is acceptable if applied consistently, but a limit must not be combined with a standard deviation as if both were standard uncertainties. This lecture uses $u = a/\sqrt{3}$ when they are combined.

EQUATION – standard uncertainty from a limit

$$u = \frac{a}{\sqrt{3}} \approx 0.58 a$$

Meaning. the standard uncertainty of a reading known only to lie within $\pm a$, all values in the range equally likely.

Variables. a the reading limit (half the smallest division, or one unit in the last digit).

Units. the unit of the quantity measured.

Interpretation. smaller than a : it is the typical error, not the extreme.

Example. metre rule: $a = 0.5 \text{ mm}$, $u = 0.29 \text{ mm}$.

Predict first

A digital stopwatch shows 0.01 s in its last digit.

Predict whether the standard uncertainty is larger or smaller than 0.01 s.

✂ TOOLKIT 2 – Classifying a source

For each source of uncertainty, ask in order:

1. **Would repeating change it?** Random (varies) or systematic (same each time).
2. **Can I estimate it from repeated readings?** Type A (use scatter) or Type B (use limit, specification, judgement).
3. **How large is it?** Scatter: $s/\sqrt{N}$. Limit $\pm a$: $a/\sqrt{3}$.
4. **Can I correct it?** If a systematic shift is known, *correct* the value and keep a (small) uncertainty on the correction.

Pendulum source	Random/systematic	Type	Estimate
Scatter of $20T$ timings	random	A	$s/\sqrt{N}$
Stopwatch resolution	random-like	B	$0.01/\sqrt{3}$ s
Length L (locating bob centre)	either	B	judgement, 0.002 m
Finite swing amplitude	systematic	B	correct, then $\pm$

IDENTIFY. A micrometer reads $+0.03$ mm with its jaws closed (zero error). It reads 5.47 mm on a rod. A thermometer reads 1.5°C in melting ice and 98.5°C in steam at 100.0°C ; it reads 36.8°C on a sample. Required: the corrected values.

SET UP. Zero error: a *constant offset*, so subtract it. Thermometer: two known points fix a straight line from reading to true value.

Prediction: the rod is slightly thinner than 5.47 mm; the thermometer reads *high* at the low end and *low* at the high end, so the sample is below 36.8°C .

EXECUTE.

Rod: $5.47 - 0.03 = 5.44$ mm.

$$\begin{aligned}\text{Thermometer: } T_{\text{true}} &= \frac{T_{\text{read}} - 1.5}{98.5 - 1.5} \times 100.0 \\ &= \frac{36.8 - 1.5}{97.0} \times 100.0 = 36.4^\circ\text{C}.\end{aligned}$$

EVALUATE. The corrections (-0.03 mm; -0.4°C) agree with the predictions. The thermometer correction is about as large as its 0.5°C reading limit, so it *cannot be ignored*. The corrections themselves carry uncertainty, which enters in Contact 2.

Common mistakes and the correct habit

Mistake

A result with no uncertainty or unit, or with more digits than the uncertainty allows (3.470 ± 0.05).

Forgetting the zero error.

Using “error” for a mistake.

Believing a small spread means an accurate result.

Repeating a measurement to remove a systematic source.

Treating the instrument resolution as the whole uncertainty.

Instead

One significant figure on δx , then round the value to match: 3.47 ± 0.05 , with the unit.

Check the zero first; subtract the offset and keep its small uncertainty.

Say *uncertainty* for the doubt and *discrepancy* for the gap to the accepted value.

Precision comes from the data; accuracy needs a reference or a second method.

Calibrate, change the method, or correct the known shift.

List every source: reaction time, judgement, resolution; the largest sets the uncertainty.

Can you now:

- say why “1.2 m” alone is an incomplete result?
- distinguish accuracy from precision on a target sketch?
- read an instrument and state its reading limit?
- say which uncertainties averaging removes, and which it does not?
- classify a source as random or systematic, Type A or Type B?

Connect: Contact 2 and PHYS 143

Repeated readings and formulas (Contact 2). In PHYS 143 every example gives data to three figures; here you learn how many of those figures are justified.

Contact 2: Quantify

How do repeated readings and formulas give a number with an uncertainty?

An opening question: ten timings

A pendulum of length $L = 0.800$ m: ten timings of 20 oscillations, t_{20} in seconds

35.92 36.05 35.88 36.11 35.97 36.03 35.94 36.08 36.00 35.96

Predict first

Sketch the ten values on a number line. (i) Estimate the best value of t_{20} . (ii) Roughly how far does a typical reading lie from it? (iii) If a colleague repeats the whole set of ten, how far would their *mean* differ from yours: about the same as the scatter of one reading, or much less?

Timing 20 oscillations rather than one is itself a design choice: the stopwatch limit is shared among 20 periods. The data are illustrative.

Decide first

1. Does the standard deviation of the readings shrink if you take a hundred of them?
2. Can the mean of ten readings be known better than any single reading?
3. Does repeating a measurement reduce a systematic uncertainty?
4. Is the uncertainty of a sum larger than the uncertainty of each term?

Decide first

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Answers

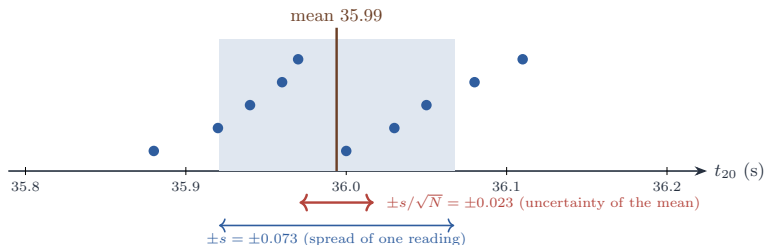
1. **No.** It settles near the true scatter of one reading; the *mean* improves.
2. **Yes**, by a factor $\sqrt{N}$ (random sources only).
3. **No.** Every reading shifts the same way.
4. **Yes, but less than the plain sum:** independent terms add in quadrature.

Defining the vocabulary of repeated measurements

Quantity	What it is	What it measures	Unit	Representation
Mean $\bar{x}$	the sum of the readings divided by N	the best estimate of the quantity	unit of x	the centre of the cloud of dots
Standard deviation s	the root-mean-square deviation about $\bar{x}$, with $N - 1$	the spread of a <i>single</i> reading	unit of x	the width of the cloud
Standard error $s/\sqrt{N}$	s divided by $\sqrt{N}$	how well the <i>mean</i> is known	unit of x	the error bar on the mean
Relative uncertainty	$\delta x/ x $	how large the doubt is compared with the value	none (%)	a percentage

Interpret: s answers *how variable is one reading?*; $s/\sqrt{N}$ answers *how well do I know the average?*

Sketch: the data and what we summarise



Rule: summarising N repeated readings $x_1, \dots, x_N$

$$\bar{x} = \frac{1}{N} \sum_i x_i, \quad s = \sqrt{\frac{\sum_i (x_i - \bar{x})^2}{N-1}}, \quad \delta\bar{x} = \frac{s}{\sqrt{N}} \quad (\text{standard error of the mean}).$$

s is the spread of a *single* reading. $s/\sqrt{N}$ is the uncertainty of the *mean*.

EQUATION – mean and standard deviation

$$\bar{x} = \frac{1}{N} \sum_i x_i, \quad s = \sqrt{\frac{\sum (x_i - \bar{x})^2}{N - 1}}$$

Meaning. the best estimate, and the spread of one reading.

Variables. x_i the N readings; $N - 1$ because $\bar{x}$ uses one degree of freedom.

Units. the unit of x .

Interpretation. about 68% of readings lie within $\bar{x} \pm s$.

Example. t_{20} : $\bar{t} = 35.994 \text{ s}$, $s = 0.073 \text{ s}$.

EQUATION – standard error of the mean

$$\delta \bar{x} = \frac{s}{\sqrt{N}}$$

Meaning. the Type A uncertainty of the mean.

Variables. s standard deviation; N number of readings.

Units. the unit of x .

Interpretation. falls as $1/\sqrt{N}$: four times the readings halve it.

Example. $0.073/\sqrt{10} = 0.023 \text{ s}$.

TOOLKIT 3 – Summarising repeated data

1. **List and sketch** the readings. Look for a pattern (drift) and for a value far from the rest.
2. **Mean** $\bar{x}$: the best estimate.
3. **Standard deviation** s with $N - 1$ in the denominator (one degree of freedom is used up by $\bar{x}$).
4. **Standard error** $s/\sqrt{N}$: the Type A uncertainty of the mean.
5. **Report** $\bar{x} \pm s/\sqrt{N}$ (Toolkit rounding), and state N .

CHECK: about two thirds of the readings lie within $\bar{x} \pm s$.

WATCH OUT: a drift (readings steadily rising) means the conditions changed. Averaging a drift is meaningless.

Worked example: the pendulum timings

IDENTIFY. $N = 10$ readings of t_{20} . Required: best value and its uncertainty, then $T = t_{20}/20$.

SET UP. Toolkit 3. *Prediction:* the mean is near 36.0 s; the readings spread by about 0.07 s.

EXECUTE.

$$\bar{t} = 359.94/10 = 35.994 \text{ s},$$

$$s = \sqrt{0.0484/9} = 0.073 \text{ s},$$

$$\frac{s}{\sqrt{N}} = \frac{0.0734}{\sqrt{10}} = 0.023 \text{ s}.$$

Resolution (Type B): $0.01/\sqrt{3} = 0.0058 \text{ s}$.
Combine the Type A and Type B terms (rule on a later slide):

$$\delta t_{20} = \sqrt{0.0232^2 + 0.0058^2} = 0.0239 \text{ s}.$$

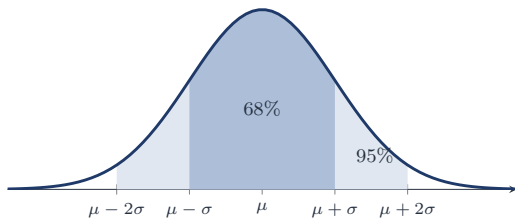
Divide by 20 (exact):

$$T = \frac{35.994}{20} = 1.7997 \text{ s},$$

$$\delta T = \frac{0.0239}{20} = 0.0012 \text{ s}.$$

EVALUATE. $\bar{t}$ and s match the predictions. The resolution term is 4 times smaller than the scatter and raises the total only from 0.0232 to 0.0239. $T = (1.7997 \pm 0.0012) \text{ s}$ (0.07%).

Why $s/\sqrt{N}$?



For scatter of the common bell-shaped (normal) kind, about 68% of single readings lie within $\bar{x} \pm s$ and about 95% within $\bar{x} \pm 2s$.

Rule: the $\sqrt{N}$ law

The standard error falls as $1/\sqrt{N}$, while s itself does *not* shrink as N grows.

N	$s/\sqrt{N}$ compared with $N = 1$
-----	------------------------------------

4	half
16	one quarter
100	one tenth

Predict first

The standard error in the pendulum example is 0.023 s. How many readings give 0.0115 s? ($4 \times 10 = 40$.) Is it worth it?

Six measurements of g (m/s^2): 9.78, 9.81, 9.75, 9.80, 9.77, 9.52

Is 9.52 a legitimate reading, or should it be discarded?

Rule: Chauvenet's criterion

Compute $\bar{x}$ and s from all N readings. For a suspect reading x_{sus} let $z = |x_{\text{sus}} - \bar{x}|/s$. The expected number of readings that far out is $N \cdot P(|Z| > z)$. If this number is **less than 0.5**, the reading may be rejected, and $\bar{x}$, s are recomputed without it.

Here $\bar{x} = 9.738$, $s = 0.109$, $z = 2.00$, so $N P = 6 \times 0.0455 = 0.27 < 0.5$: **reject**. Without it: $\bar{x} = 9.782$, $s = 0.024$, $s/\sqrt{5} = 0.011$, so $g = (9.78 \pm 0.01) \text{m/s}^2$.

Caution

A statistical test is not enough. Look for a cause (misread, bump, wrong start), *record* the rejection and report that you made it. Never discard a reading only because it is inconvenient.

Rule: independent contributions combine in quadrature

If a result depends on several *independent* sources, with standard uncertainties $u_1, u_2, \dots$ in the same unit, the combined standard uncertainty is

$$u_c = \sqrt{u_1^2 + u_2^2 + \dots}$$

Justification on the next slides (and in the supplementary note).

Physical meaning

The largest term dominates: $\sqrt{0.0232^2 + 0.0058^2} = 0.0239$ barely differs from 0.0232. **It is pointless to improve a small source while a large one remains.** This is the single most useful habit in laboratory work.

An opening question: how does δL and δT become δg ?

Pendulum formula

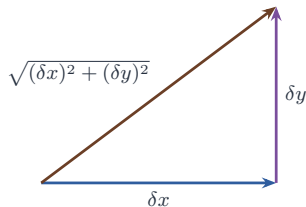
$$g = \frac{4\pi^2 L}{T^2}, \quad L = 0.800 \pm 0.002 \text{ m}, \quad T = 1.795 \pm 0.005 \text{ s}.$$

Predict first

Percentage uncertainty of L ? Of T ? Because T is *squared*, does its uncertainty count once, twice or half? Which quantity do you expect to dominate δg ? Is $\delta g/g$ the sum of the percentages, or smaller?

We return to this after the rules: the answer is a worked example.

Why quadrature?



independent errors rarely both reach their worst value together, so the combined size is smaller

 $\delta x + \delta y$: both at their worst

Two independent sources δx and δy would give the maximum combined uncertainty $\delta x + \delta y$ only if both were at their worst at once, in the same direction.

Independent errors are sometimes positive, sometimes negative, so they partly cancel. Their combined size behaves like the hypotenuse of a right triangle. The simple sum is an *upper bound*.

Rule: propagation for common operations (independent uncertainties)

Sum or difference $q = x \pm y$ $\delta q = \sqrt{(\delta x)^2 + (\delta y)^2}$ (*absolute* uncertainties)

Product or quotient $q = xy$
or x/y $\frac{\delta q}{|q|} = \sqrt{\left(\frac{\delta x}{|x|}\right)^2 + \left(\frac{\delta y}{|y|}\right)^2}$ (*relative*)

Power $q = x^n$ $\frac{\delta q}{|q|} = |n| \frac{\delta x}{|x|}$

Constant factor $q = Bx$ $\delta q = |B| \delta x$ (exact B ; for example $\frac{1}{2}$, $4\pi^2$, 2)

Caution

The same quantity appearing twice is *not* independent: treat $x \cdot x$ as the power x^2 , giving $2 \delta x/x$, not $\sqrt{2} \delta x/x$. A difference of two nearly equal numbers has a large *relative* uncertainty.

Rule: propagation for any function $q = f(x, y, \dots)$ (independent uncertainties)

$$\delta q = \sqrt{\left(\frac{\partial f}{\partial x} \delta x\right)^2 + \left(\frac{\partial f}{\partial y} \delta y\right)^2 + \dots}$$

Physical meaning

The term $(\partial f / \partial x) \delta x$ is the change in q produced by a change δx in x alone, with every other variable held fixed: the *contribution* of x . All four special cases are this formula applied to $x \pm y$, xy , x/y and x^n .

Partial derivative: differentiate with respect to one variable, treating the others as constants. For $f = L \sin \theta$: $\partial f / \partial L = \sin \theta$ and $\partial f / \partial \theta = L \cos \theta$. Angles must be in radians.

EQUATION – quadrature

$$u_c = \sqrt{u_1^2 + u_2^2 + \dots}$$

Meaning. the combined uncertainty of independent sources.

Variables. u_i the standard uncertainties, in the same unit.

Units. the unit of the result.

Interpretation. the largest term dominates; the plain sum is an upper bound.

Example. $\sqrt{0.0232^2 + 0.0058^2} = 0.0239$ s.

EQUATION – product, quotient and power

$$\frac{\delta q}{|q|} = \sqrt{\left(\frac{\delta x}{|x|}\right)^2 + \left(\frac{\delta y}{|y|}\right)^2}, \quad \frac{\delta q}{|q|} = |n| \frac{\delta x}{|x|}$$

Meaning. relative uncertainties combine for xy , x/y ; a power multiplies the relative uncertainty by $|n|$.

Variables. $q = xy$, x/y , x^n .

Units. none (relative).

Interpretation. a squared quantity counts twice.

Example. $g = 4\pi^2 L/T^2$: $\delta g/g = 0.61\%$.

TOOLKIT 4 – Propagating uncertainty

1. **Write the formula** for the result. Check that the inputs are independent.
2. **Choose the rule:** sums and differences (absolute), products and powers (relative), anything else (partial derivatives).
3. **Compute each contribution separately** and tabulate them.
4. **Combine in quadrature.**
5. **Find the dominant contribution** (the largest). A contribution smaller than about a third of it hardly matters.
6. **Round** by Toolkit 1 conventions: one significant figure on δq , then the value.

CHECK: δq must be at least as large as the largest single contribution and at most their plain sum.

Lecturer example: g from the pendulum (relative-uncertainty rule)

IDENTIFY. $L = 0.800 \pm 0.002$ m,
 $T = 1.795 \pm 0.005$ s. Required: $g = 4\pi^2 L/T^2$
and δg .

SET UP. A product, a quotient and a power:
use relative uncertainties. *Prediction:*
 $g \approx 9.8$ m/s²; the squared T counts double.

EXECUTE.

$$g = \frac{4\pi^2(0.800)}{1.795^2} = 9.80 \text{ m/s}^2,$$
$$\frac{\delta L}{L} = \frac{0.002}{0.800} = 0.25\%,$$
$$2\frac{\delta T}{T} = 2 \times \frac{0.005}{1.795} = 0.56\%.$$

$$\frac{\delta g}{g} = \sqrt{0.25^2 + 0.56^2}\% = 0.61\%,$$

$$\delta g = 9.80 \times 0.0061 = 0.06 \text{ m/s}^2.$$

EVALUATE. $g = (9.80 \pm 0.06) \text{ m/s}^2$.

- T dominates: its 0.28% doubles to 0.56%.
- Check: 0.61% lies between 0.56% (largest) and 0.81% (plain sum). ✓
- Value agrees with 9.78: $D = +0.02 < 0.06$.
- The tool to improve: time more oscillations.

(a) Temperature rise. $T_i = 21.3 \pm 0.2^\circ\text{C}$,
 $T_f = 34.6 \pm 0.2^\circ\text{C}$.

Difference: absolute uncertainties in
quadrature:

$$\Delta T = 13.3^\circ\text{C},$$

$$\delta\Delta T = \sqrt{0.2^2 + 0.2^2} = 0.28^\circ\text{C}.$$

$\Delta T = (13.3 \pm 0.3)^\circ\text{C}$ (the plain sum 0.4 is
the upper bound). The relative uncertainty
has grown: $0.2/34.6 = 0.6\%$ became
 $0.28/13.3 = 2.1\%$.

(b) A general formula. A ramp has length
 $L = 1.200 \pm 0.002\text{ m}$ at angle $\theta = 30.0^\circ \pm 0.5^\circ$.
Height $h = L \sin \theta$.

$h = 1.200 \sin 30.0^\circ = 0.600\text{ m}$. With
 $\delta\theta = 0.5^\circ = 0.00873\text{ rad}$:

$$\sin \theta \delta L = 0.0010,$$

$$L \cos \theta \delta\theta = 0.0091,$$

$$\begin{aligned}\delta h &= \sqrt{0.0010^2 + 0.0091^2} \\ &= 0.009\text{ m}.\end{aligned}$$

$h = (0.600 \pm 0.009)\text{ m}$; the angle dominates.

Independent problem: select the rule, predict, then calculate

A brass cylinder has mass $m = 95.2 \pm 0.1$ g, diameter $d = 1.250 \pm 0.005$ cm and height $h = 8.45 \pm 0.02$ cm. The density is $\rho = \frac{m}{V} = \frac{4m}{\pi d^2 h}$.

1. Calculate ρ and $\delta\rho$ and write the result in full.
2. Which of the three measurements should be improved first? By how much would $\delta\rho$ fall if its relative uncertainty were halved?

Rule: product, quotient, power: relative uncertainties. *Prediction:* d enters squared and has the largest relative uncertainty (0.4%), so it dominates.

$$\rho = \frac{4(95.2)}{\pi(1.250)^2(8.45)} = 9.18 \text{ g/cm}^3.$$

Source	Relative contribution	Value
m	$0.1/95.2$	0.105%
d (squared)	$2 \times 0.005/1.250$	0.80%
h	$0.02/8.45$	0.24%

$$\frac{\delta\rho}{\rho} = \sqrt{0.105^2 + 0.80^2 + 0.24^2}\% = 0.84\%, \quad \delta\rho = 0.077 \rightarrow 0.08 \text{ g/cm}^3.$$

$\rho = (9.18 \pm 0.08) \text{ g/cm}^3$. The diameter dominates (about 90% of the variance). Halving $\delta d/d$ gives $\sqrt{0.105^2 + 0.40^2 + 0.24^2} = 0.48\%$, so $\delta\rho \approx 0.04 \text{ g/cm}^3$.

First-level rules (OpenStax §1.6)

- Multiplying: keep the fewest significant figures.
- Adding: keep the fewest decimal places.
- Quick, needs no uncertainties, but *hides* what dominates and can mislead.

Propagation

- Uses the actual δx of each quantity.
- Reveals the dominant source.
- Is the standard for any reported laboratory result.

Example where the shortcut fails

$\rho = 9.18 \text{ g/cm}^3$ has three significant figures from the density example. Its real uncertainty is 0.08, so the second decimal is already doubtful: the sig-fig rule alone could not say this. A difference $34.6 - 21.3 = 13.3$ also shows ΔT with three digits while its relative uncertainty is 2%.

Working habit: carry two or three extra digits through the calculation and round once, at the end, using the uncertainty.

Common mistakes and the correct habit

Mistake

Quoting s where $s/\sqrt{N}$ is meant, or dividing by N instead of $N - 1$.

Averaging a drifting quantity, or discarding an outlier silently.

Adding uncertainties linearly, or adding relative uncertainties for a sum.

Forgetting the exponent: T^2 counts double.

Treating $x \cdot x$ as two independent quantities.

Rounding in the middle, or using degrees where radians are needed.

Instead

s is the spread of one reading; $s/\sqrt{N}$ is the uncertainty of the mean; use $N - 1$.

Plot the data; reject only with a cause and a criterion, and report it.

Quadrature for independent sources; absolute for sums, relative for products.

Multiply the relative uncertainty by $|n|$ for x^n .

Write it as the power x^2 .

Carry extra digits and round once at the end; radians in derivatives.

Can you now:

- calculate $\bar{x}$, s and $s/\sqrt{N}$, and say what each means?
- decide about an outlier, and report the decision?
- combine independent uncertainties in quadrature?
- propagate through sums, products, powers and a general formula?
- find the dominant source and say how to improve the result?

Connect: Contact 3 and PHYS 143

When a quantity is varied deliberately we fit a graph and need the uncertainty of the slope. The same propagation rules give g (Lecture 2), k (Lecture 4) and c (Lecture 9).

Contact 3: Apply

Graphs, budgets, comparison with theory and the report

An opening question: a graph in place of one number

Hooke's law: hang masses on a spring, measure the length l with a ruler (limit 0.1 cm on each reading)

m (g)	50	100	150	200	250	300	350	400
l (cm)	14.1	16.0	17.9	20.1	21.8	23.8	25.9	27.6

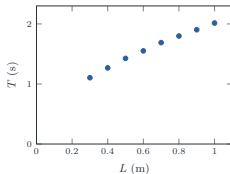
Predict first

Sketch l against m . Is it a straight line through the origin? What do the slope and the intercept mean physically? Would a single measurement of k from one mass be as trustworthy as the fit?

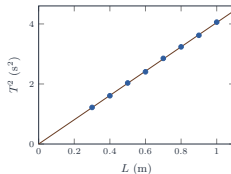
The model: $l = l_0 + (g/k) m$, with l_0 the unstretched length and k the spring constant. The data are illustrative.

Toolkit 5: linearising and fitting

(a) T against L : a curve



(b) T^2 against L : a straight line



TOOLKIT 5 – Linearise, then fit

1. Write the model; rearrange to $y = mx + c$.
2. Plot y against x with error bars.
3. Fit; read slope, intercept and their uncertainties.
4. Convert the slope to the physical quantity (propagate).

Relation	Plot	Slope
$T^2 = (4\pi^2/g) L$	T^2 against L	$4\pi^2/g$
$l = l_0 + (g/k) m$	l against m	g/k
$P = nRT (1/V)$	P against $1/V$	nRT

- An **error bar** of length $\pm\sigma_y$ on each point shows the uncertainty of that reading.
- About two thirds of the bars should touch the line: a line missing all bars, or touching all with room to spare, signals trouble.
- **By hand:** draw the best line, then the steepest and shallowest lines consistent with the bars. The slope range gives δm . Good for checking.
- **By calculation:** least squares (next slide). It chooses the line that minimises the sum of squared vertical distances.

Why a fit helps

Each point carries a small uncertainty. The line uses *all* the points, so its slope is better determined than any single ratio l/m . It also reveals an offset (the intercept) that a single reading would hide.

Decide first

1. Must the best line pass through every point?
2. Is the slope of a fitted line better known than any single ratio y/x ?
3. Does a small χ^2_{red} (say 0.1) prove the data are excellent?
4. If the line fits well, is the physical model proved?

Decide first

1. Must the best line pass through every point?
2. Is the slope of a fitted line better known than any single ratio y/x ?
3. Does a small χ^2_{red} (say 0.1) prove the data are excellent?
4. If the line fits well, is the physical model proved?

Answers

1. **No.** It passes within the error bars of about two thirds of the points.
2. **Yes.** The line uses every point.
3. **No.** It suggests the uncertainties were overestimated.
4. **No.** A good fit supports the model; it cannot prove it.

Defining the vocabulary of fitting and comparison

Quantity	What it is	What it measures	Unit	Representation
Slope m and intercept c	the parameters of the best straight line	the rate of change of y with x ; the value of y at $x = 0$	unit of y /unit of x ; unit of y	the line drawn through the points
Residual	$y_i - mx_i - c$	how far a point lies from the line	unit of y	the vertical gap
Reduced χ^2	$\chi^2/(N - 2)$	whether the line fits within the stated uncertainties	none	a number near 1 if the fit is good
Number n of combined uncertainties	$ a - b /\sqrt{(\delta a)^2 + (\delta b)^2}$	how far apart two results are	none	the gap in units of the uncertainty

Interpret: the residual answers *does the model miss a pattern in the data?*; n answers *do these two results agree?*

Rule: least-squares line $y = mx + c$, equal uncertainty σ_y on every y

$$\Delta = N \sum x^2 - \left(\sum x \right)^2, \quad m = \frac{N \sum xy - \sum x \sum y}{\Delta}, \quad c = \frac{\sum x^2 \sum y - \sum x \sum xy}{\Delta}.$$

$$\delta m = \sigma_y \sqrt{\frac{N}{\Delta}}, \quad \delta c = \sigma_y \sqrt{\frac{\sum x^2}{\Delta}}.$$

If σ_y is unknown, estimate it from the scatter about the line:

$$\sigma_y \approx s_y = \sqrt{\frac{\sum (y_i - mx_i - c)^2}{N - 2}}.$$

Assumptions: the uncertainty is in y only (the x values are much better known), is the same for every point, and the points are independent. Weighted fits, when the σ_i differ, are in the supplementary note.

EQUATION – slope and intercept

$$m = \frac{N \sum xy - \sum x \sum y}{\Delta}, \quad \delta m = \sigma_y \sqrt{\frac{N}{\Delta}}$$

Meaning. the slope that minimises the sum of the squared vertical distances to the points, and its uncertainty.

Variables. $\Delta = N \sum x^2 - (\sum x)^2$; σ_y the uncertainty of each y .

Units. slope: unit of y per unit of x .

Interpretation. uses every point, so it is better known than any one ratio.

Example. the spring: $0.0389 \pm 0.0003 \text{ cm/g}$.

Predict first

Doubling the number of equally spaced points: predict by what factor the slope uncertainty changes.

Worked example: the spring

IDENTIFY. $N = 8$ points (m, l) ; $\sigma_l = 0.1$ cm.

Required: slope, intercept, k .

SET UP. Line $l = c + (\text{slope}) m$. *Prediction:*
slope near 0.04 cm/g (about 2 cm per 50 g);
intercept near 12 cm.

EXECUTE. $\sum m = 1800$ g,

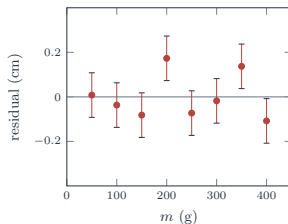
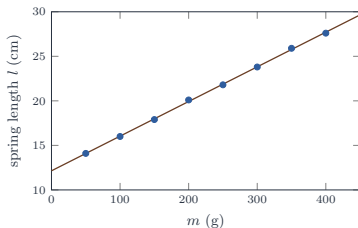
$$\sum m^2 = 5.10 \times 10^5 \text{ g}^2,$$

$$\Delta = 8(5.10 \times 10^5) - 1800^2 = 8.4 \times 10^5 \text{ g}^2.$$

The formulas give

$$\text{slope} = (0.0389 \pm 0.0003) \text{ cm/g},$$

$$c = (12.15 \pm 0.08) \text{ cm}.$$



EVALUATE. Predictions agree. The line passes near all bars (left). The intercept is the unstretched length, 12.15 ± 0.08 cm. The residuals (right) scatter about zero with no pattern.

Rule: reduced chi-squared

$$\chi^2 = \sum_i \left(\frac{y_i - mx_i - c}{\sigma_i} \right)^2, \quad \nu = N - 2, \quad \chi_{\text{red}}^2 = \frac{\chi^2}{\nu}.$$

Rule of thumb: $\chi_{\text{red}}^2 \approx 1$: the model fits within the stated uncertainties. $\chi_{\text{red}}^2 \gg 1$: the model is wrong, or the σ_i are too small. $\chi_{\text{red}}^2 \ll 1$: the σ_i are too large, or the data have been fitted too well.

Spring: $\chi^2 = 7.4$, $\nu = 6$, $\chi_{\text{red}}^2 = 1.23$. Acceptable (the probability of exceeding this value by chance is 0.29).

Residual plot: always draw it. A random scatter about zero supports the model; a curve or a trend means the model is wrong, even if χ_{red}^2 looks fine.

EQUATION – goodness of fit

$$\chi^2 = \sum_i \left(\frac{y_i - mx_i - c}{\sigma_i} \right)^2, \quad \chi^2_{\text{red}} = \frac{\chi^2}{N - 2}$$

Meaning. the squared residuals, each in units of its own uncertainty.

Variables. σ_i the uncertainty of point i ; $N - 2$ degrees of freedom.

Units. none.

Interpretation. near 1: fits within the uncertainties;
 $\gg 1$: wrong model or underestimated σ ; $\ll 1$: overestimated σ .

Example. the spring: 1.23.

Predict first

Predict what happens to χ^2_{red} if every error bar is doubled.

IDENTIFY. Slope = g/k with $g = 9.78 \text{ m/s}^2$ (uncertainty negligible). Required: k .

SET UP. Convert the slope to SI:

$1 \text{ cm/g} = 10 \text{ m/kg}$, so

$0.0389 \text{ cm/g} = 0.389 \text{ m/kg}$. $k = g/\text{slope}$ is a quotient with an exact numerator, so $\delta k/k = \delta(\text{slope})/\text{slope} = 0.0003/0.0389 = 0.8\%$.

EXECUTE.

$$k = \frac{9.78}{0.389} = 25.1 \text{ N/m},$$

$$\delta k = 0.008 \times 25.1 = 0.2 \text{ N/m}.$$

EVALUATE. $k = (25.1 \pm 0.2) \text{ N/m}$. Units: $(\text{m/s}^2)/(\text{m/kg}) = \text{kg/s}^2 = \text{N/m}$. ✓

Compare two methods

A second experiment, timing the oscillations, gives $k = 24.6 \pm 0.3 \text{ N/m}$. The difference is 0.5; the combined uncertainty is $\sqrt{0.2^2 + 0.3^2} = 0.36$, so the difference is 1.4 combined uncertainties: the two methods agree (rule on a later slide).

TOOLKIT 6 – Building an uncertainty budget

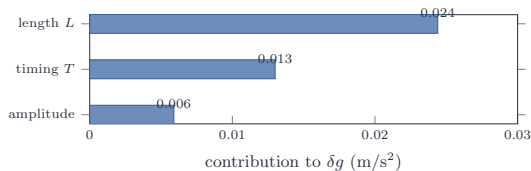
A table with one row per source, filled *before* any final claim:

1. **Source**, with its Type (A or B) and whether random or systematic.
2. **Standard uncertainty** of the input, from Toolkit 2.
3. **Contribution to the result**: $|\partial f / \partial x| \delta x$ (or the relative rule).
4. **Share of the total variance**: $(\text{contribution})^2 / (\delta q)^2$.
5. **Total** in quadrature, the **dominant source**, and what to do about it.

CHECK: systematic effects are *corrected* first, then their small remaining uncertainty enters as a row.

The whole experiment: g from the pendulum

Source	Type	Input	Contribution to g	Variance share
Length $L = 0.800$ m	B	0.002 m	0.024	75%
Period $T = 1.7997$ s	A, B	0.0012 s	0.013	21%
Amplitude correction ($8^\circ \pm 1^\circ$)	B	$\pm 1^\circ$	0.006	4%
Total (quadrature)			0.028	100%



Reading the budget. The *length* dominates. The uncertainty of T from ten timings is already small; further repeats would help little. To improve the result, measure L better: a longer rule, or a better way to locate the bob centre.

A systematic correction: finite amplitude

IDENTIFY. The formula $g = 4\pi^2 L/T^2$ assumes small swings. The measured period at amplitude θ_0 is longer: $T = T_0 (1 + \theta_0^2/16)$ (first correction, θ_0 in radians). So g from the raw T is too small.

SET UP. $\theta_0 = 8^\circ = 0.1396$ rad. *Prediction:* the correction is a fraction of one percent, raising g slightly.

EXECUTE. Raw:

$$g_{\text{raw}} = 4\pi^2(0.800)/1.7997^2 = 9.751 \text{ m/s}^2.$$

$$\text{Correction factor } (1 + \theta_0^2/16)^2 = 1.00244:$$

$$g = 9.775 \text{ m/s}^2. \text{ Uncertainty of the correction (from } \pm 1^\circ \text{ in } \theta_0): 0.006.$$

EVALUATE. Combined with the other two rows:

$$\delta g = \sqrt{0.024^2 + 0.013^2 + 0.006^2} = 0.028.$$

$$g = (9.77 \pm 0.03) \text{ m/s}^2$$

- The correction (+0.024) is about as large as the total uncertainty, so it *must* be applied and reported.
- Without it the raw 9.75 ± 0.03 would still agree with 9.78, but only because the budget is generous.

Rule: comparing two results (or a result and an accepted value)

For results $a \pm \delta a$ and $b \pm \delta b$ define

$$n = \frac{|a - b|}{\sqrt{(\delta a)^2 + (\delta b)^2}} \quad (\delta b = 0 \text{ if } b \text{ is an accepted value}).$$

$n \lesssim 2$: **consistent**. $2 \lesssim n \lesssim 3$: **doubtful**; look harder. $n \gtrsim 3$: **inconsistent**; a cause is likely.

Comparison	n	Verdict
Pendulum: 9.77 ± 0.03 against accepted 9.78	$0.01/0.03 = 0.3$	consistent
Spring: static 25.1 ± 0.2 against dynamic 24.6 ± 0.3	$0.5/0.36 = 1.4$	consistent
Student B: 9.62 ± 0.04 against accepted 9.78	$0.16/0.04 = 4$	inconsistent

Caution. Agreement does not prove a theory right; disagreement does not prove it wrong. Look first for a missing source, an underestimated uncertainty or a false assumption.

EQUATION – the n -test

$$n = \frac{|a - b|}{\sqrt{(\delta a)^2 + (\delta b)^2}}$$

Meaning. the distance between two results in units of their combined uncertainty.

Variables. $a \pm \delta a$ and $b \pm \delta b$; $\delta b = 0$ for an accepted value.

Units. none.

Interpretation. $n \lesssim 2$ consistent; $n \gtrsim 3$ inconsistent.

Example. spring, static against dynamic: $n = 1.4$.

Predict first

9.62 ± 0.04 against 9.78 :
predict whether the two agree before computing n .

Plan a better pendulum measurement

The target is $\delta g \leq 0.015 \text{ m/s}^2$ (half of today's 0.028). Using the budget, what should be improved, and by how much?

Reasoning. L contributes 0.024: halving δL to 0.001 m gives 0.012. T contributes 0.013: 40 timings in place of 10 halve it to 0.006. The amplitude term (0.006) stays.

$$\begin{aligned}\delta g &= \sqrt{0.012^2 + 0.006^2 + 0.006^2} \\ &= 0.015 \text{ m/s}^2.\end{aligned}$$

Lesson

- Improving T alone leaves $\sqrt{0.024^2 + 0.006^2 + 0.006^2} = 0.025$: almost nothing gained.
- Improving L alone gives $\sqrt{0.012^2 + 0.013^2 + 0.006^2} = 0.019$: not enough.
- Both are needed. The budget tells you *before* the experiment is repeated.

Application problem: specific heat capacity of a metal block

A metal block of mass 0.150 ± 0.001 kg is heated to 98.0 ± 0.5 °C and dropped into 0.200 ± 0.001 kg of water at 21.3 ± 0.2 °C in an insulated container. The final temperature is 30.5 ± 0.2 °C. Take $c_{\text{water}} = 4186$ J/(kg K) (exact) and neglect the container.

1. Predict whether the block's temperature fall is larger or smaller than the water's temperature rise, and why.
2. Find c for the metal and its uncertainty. Which measurement dominates?
3. The accepted value for aluminium is 900 J/(kg K). Is the block aluminium? State any systematic effect you suspect.

Use IDENTIFY, SET UP, EXECUTE, EVALUATE. Choose the rules yourself.

IDENTIFY and SET UP. Heat lost by the block equals heat gained by the water, $c m_s (T_s - T_f) = m_w c_w (T_f - T_w)$, so

$$c = \frac{m_w c_w (T_f - T_w)}{m_s (T_s - T_f)}.$$

Prediction: the block's mass is smaller and c is smaller than the water's, so its temperature change is larger: about 67.5°C against 9.2°C .

EXECUTE. $c = \frac{0.200(4186)(9.2)}{0.150(67.5)} = 761 \text{ J/(kg K)}$. Contributions (each variable in turn, others fixed): m_w : 4; T_w : 17; m_s : 5; T_s : 6; T_f : 19 (all in J/(kg K)). Total in quadrature = 26.

EVALUATE. $c = (760 \pm 30) \text{ J/(kg K)}$. The final temperature (0.2 K on a 9.2 K rise) and the water temperature dominate. Against 900: $n = 139/26 \approx 5$: **inconsistent**. A systematic cause: heat lost to the container and to the air makes the water seem to gain less heat, giving a low c . The block may be aluminium, but a better experiment (insulate, include the calorimeter's heat capacity, larger temperature rise) is needed to say.

Reporting the result in the laboratory report

A usual structure (confirm the format your course requires):

1. Title, aim, theory (formula and assumptions).
2. Apparatus, with each instrument's resolution.
3. Method, including steps taken to reduce uncertainty.
4. **Data** in tables: quantity, unit and uncertainty in each column heading.
5. **Analysis**: sample calculation, propagation, graph with error bars, fit parameters.
6. **Result** in full reporting form with percent uncertainty.
7. **Discussion**: agreement (n), budget, dominant source, systematic effects, improvements.
8. Conclusion.

✓ Checklist before handing in

- Every number has a unit.
- Every result has an uncertainty, rounded correctly.
- Systematic effects were looked for and stated.
- Graph: labelled axes with units, error bars, fitted line, residuals.
- The budget names the dominant source.
- The comparison states n , not just “close”.
- The conclusion says what was learned, not only the number.

Common mistakes and the correct habit

Mistake

Fitting without error bars or a residual plot.

Forcing the line through the origin when the model has an intercept.

Ignoring a large uncertainty in x when the fit assumes y only.

Reading a good fit as proof of the model, or a large n as proof it is wrong.

Omitting the budget, so the dominant source is never named.

Blaming “human error” or skipping the comparison.

Instead

Always draw both; a trend in the residuals means the model is wrong.

Fit $y = mx + c$ and read the intercept physically.

Put the better-known quantity on the x axis, or use a method that includes both.

Look first for a missing source, an underestimated uncertainty or a false assumption.

Tabulate every source and its variance share before any claim.

Estimate each source quantitatively and report n .

Every experiment in PHYS 105

Momentum, motion, force, circular and rotational motion, gravitation and heat all end with a result, an uncertainty and a comparison.

PHYS 143

The pendulum and spring experiments are the laboratory counterparts of Lectures 2 and 4; specific heat (Lecture 9) is the calorimetry problem above.

Beyond the laboratory

Any claim from data, in engineering, medicine or climate science, is only as good as its stated uncertainty.

✓ Laboratory habits from this lecture

Write the uncertainty beside each reading, at the bench. Look for systematics before you repeat. Find the dominant source early and spend your effort there. Round last. Compare in terms of n .

Key ideas

- A result is a best value, an uncertainty and a unit.
- Averaging reduces random uncertainty by $\sqrt{N}$, not systematic.
- Independent uncertainties add in quadrature; the largest dominates.
- A squared quantity counts twice in the relative uncertainty.
- A straight-line fit uses every point; check the residuals and χ^2_{red} .
- Compare results with n ; correct systematics first.

Where to look

Definitions: the vocabulary tables of each contact.

Common mistakes: the last slide of each contact.

Connections: the Connect boxes and slide.

Summary: core equations

Idea	Equation	Conditions and units
Result	$x = (x_{\text{best}} \pm \delta x) \text{ unit}$	one standard uncertainty
Reading limit	$u = a/\sqrt{3}$	rectangular distribution
Repeated readings	$\bar{x}, \quad s = \sqrt{\sum (x_i - \bar{x})^2 / (N - 1)}, \quad \delta \bar{x} = s/\sqrt{N}$	unit of x
Quadrature	$u_c = \sqrt{u_1^2 + u_2^2 + \dots}$	independent sources
Products, powers	$\delta q/ q = \sqrt{(\delta x/x)^2 + (\delta y/y)^2}, \quad n \delta x/ x $	relative
General	$\delta q = \sqrt{\sum (\partial f / \partial x_i \delta x_i)^2}$	radians
Least squares	$m = (N \sum xy - \sum x \sum y) / \Delta, \quad \delta m = \sigma_y \sqrt{N/\Delta}$	y uncertainty only
Fit quality	$\chi_{\text{red}}^2 = \chi^2 / (N - 2)$	near 1
Comparison	$n = a - b / \sqrt{(\delta a)^2 + (\delta b)^2}$	$n \lesssim 2$ consistent

I can ...

1. report a result as (value $\pm$ uncertainty) unit with correct rounding and percent uncertainty;
2. distinguish accuracy from precision, random from systematic, Type A from Type B;
3. calculate $\bar{x}$, s and $s/\sqrt{N}$, and judge an outlier;
4. propagate uncertainty through sums, products, powers and general formulas;
5. fit a straight line and interpret slope, intercept, residuals and χ^2_{red} ;
6. build a budget, test agreement with n , and write the report.

Toolkits this lecture

- T1 Reading an instrument
- T2 Classifying a source
- T3 Summarising repeated data
- T4 Propagating uncertainty
- T5 Linearising and fitting
- T6 Building a budget and comparing results

Next: the tutorial sheet (answers at the end) and the supplementary note (derivations, tables, weighted fits).